



StoreFlow AI

StoreFlow AI: Transforming In-Store Video Feeds into Real-Time Spatial Intelligence

A Privacy-Preserving, Ultra-Low-Cost Computer Vision Engine for Physical Retail

// SYSTEM OVERVIEW & THESIS

Blind Retail Floor Analysis

Physical retailers generate petabytes of raw video daily, yet remain functionally blind to real-time customer behavior, shelf interaction, and labor bottlenecks due to severe compute costs.

// CRITICAL FAILURE POINTS

- **Prohibitive Cloud Egress:** Streaming high-res video triggers massive bandwidth costs.
- **Privacy Vulnerabilities:** Storing raw biometric video creates compliance hazards.
- **Heavy Edge Hardware:** High CapEx GPU servers per store eliminate ROI.

// 4-STAGE OPERATING LOOP

1. OBSERVE

Zero-copy edge stream parsing directly from existing CCTV cameras.



2. DIAGNOSE

Privacy-first vectorization of spatial movement and dwell events.



3. PRESCRIBE

Real-time automated floor alerts & queue mitigation triggers.



4. VERIFY

Continuous closed-loop attribution of labor allocation and sales uplift.

\$0

NEW CAPEX (USES EXISTING RTSP STREAMS)

14,000x

BANDWIDTH REDUCTION VIA VECTOR EXTRACTION

>97%

TCO REDUCTION VS. CLOUD CV ALTERNATIVES

Instant retro-fit on existing infra!

System Architecture: Edge-Native Zero-Copy Computer Vision Pipeline

✓ StoreFlow AI

Our system transforms oblique RTSP camera feeds into metric 2D store intelligence through a highly robust, edge-native pipeline built for commodity hardware. The practical stack utilizes OpenCV and GStreamer, feeding into Ultralytics YOLOv10/v8, ByteTrack/OC-SORT, and Shapely, ultimately delivering data via MQTT/WebSockets to a React/WebGL frontend.

INGESTION & ZERO-COPY MEMORY

Hardware zero-copy mechanisms process 8–16 heterogeneous RTSP feeds via GStreamer/OpenCV hardware decoding (`nvv4l2decoder` / `vaapicodecbin`). By keeping NV12 frames in unified DMA memory (`/dev/dma_buf`), CPU memory bottlenecks are completely eliminated.

INFERENCE, SPATIAL & TRANSPORT

- **Edge Stack:** INT8-quantized YOLOv10-Nano with NMS-free dual label assignment fused with OC-SORT for dense, non-linear tracking.
- **Spatial Metric:** Metric homography and disjoint camera handoff project ground-contact points to metric store coordinates; tracklets fused via face-masked OSNet-0.5x embeddings.
- **Transport & Resilience:** Publishes anonymous trajectory JSON tokens (65 bytes) over MQTT/TLS. Local NVMe SSD spools up to 90 days of trajectory data during outages.

TIER 1: INGESTION LAYER (RTSP FEEDS)

Hardware Decoding (`nvv4l2decoder` / `vaapicodecbin`) | 8–16 Heterogeneous RTSP Feeds | Unified DMA Memory (`/dev/dma_buf`)

TIER 2: EDGE APPLIANCE (INFERENCE & TRACKING)

INT8 YOLOv10-Nano (NMS-Free) | OC-SORT
Face-Masked OSNet-0.5x Embeddings

⚡ 1.84ms Inference Latency

TIER 3: SPATIAL ANALYTICS HUB

Disjoint Camera Handoff | Anonymous Trajectory Tokens (65 bytes) | Local NVMe SSD Spooling (Up to 90 Days)

TIER 4: EXECUTIVE COCKPIT (FRONTEND)

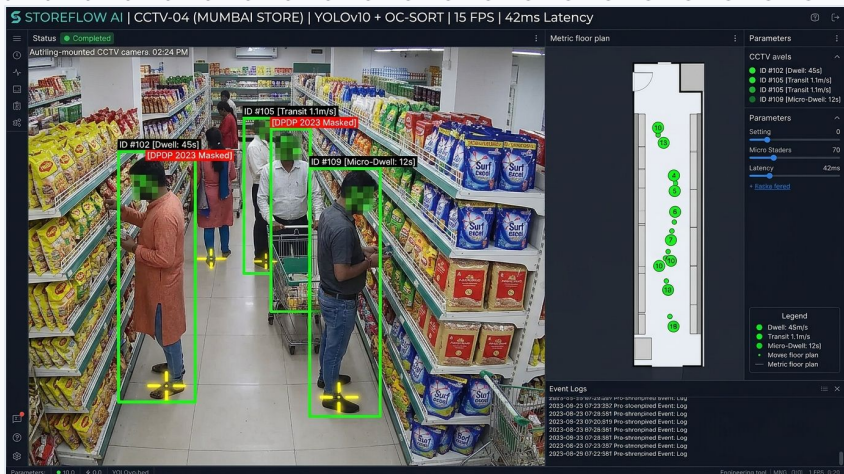
MQTT / TLS & WebSockets Data Delivery | React & WebGL 2D Metric Store Intelligence Rendering

🔒 Zero video saved to disk (<3.5 KB/s)

Computer Vision Deep-Dive: Metric BEV & Camera Fusion



StoreFlow AI



< 0.11m

CAD Ground Accuracy
Ankle regression vs 2.0m error

86.4%

Cross-Cam IDF1 Score
Velocity-constrained CLM

Parallax-free ground projection via ankle keypoints!

01 / PARALLAX-FREE GROUND CONTACT

Standard centroid projection causes **1.2–2.0m parallax error** under 45° tilt. StoreFlow locates ground contact points via **ankle-keypoint regression** to eliminate lower-body occlusion bias.

Achieves **< 0.11m metric accuracy** mapped directly onto store CAD layout coordinates.

02 / ZERO-TOUCH VANISHING POINT CALIBRATION

Automated calibration intersects parallel gondola aisle lines and orthogonal floor tile lines. Enforces an **absolute conic orthogonality constraint** to compute focal length & tilt angle without manual targets.

Formulates metric homography $H \in \mathbb{R}^{3 \times 3}$ to map 2D image coordinates to 3D store floor coordinates.

03 / FACE-MASKED RE-ID & DISJOINT CLM

Zero-PII Architecture: Upper 25% of bounding boxes are zero-masked in RAM prior to extraction. **OSNet-0.5x** generates non-reversible 128-d embeddings capturing outerwear color & geometry.

Spatio-Temporal Camera Link Model (CLM) constrains disjoint matching by human walking physics (**0.8–2.5 m/s**) along aisle geodesics. Prunes impossible links in **O(1) time** for an **86.4% IDF1 score**.

Spatial Intelligence: Physics-Based Bottleneck Detection

SHELF DWELL ANALYTICS & FLUID DYNAMICS ENGINE

STAGE 01 // SHELF DWELL & VORONOI FRONTAGES

High-Precision Dwell Engine: Filters velocity > 1.1 m/s to isolate true browsing from transit.

Voronoi Grid & Smoothing: Savitzky-Golay trajectory smoothing mapped across 1.0m micro-frontages.

EPF = (Dwell > 1.1 m/s Excluded) / Total Frontage Passby

STAGE 02 // FLUID MECHANICS & STORE FRICTION INDEX

Accumulation Detection: Triggered when velocity divergence $\nabla \cdot \mathbf{v} < 0$ and density exceeds **1.2 persons/m²**.

Group Deflection Filter: Removes family/group cluster false positives.

MST Queue Orientation: Minimum Spanning Tree maps checkout queue angles to flag spillover within **35 seconds**.

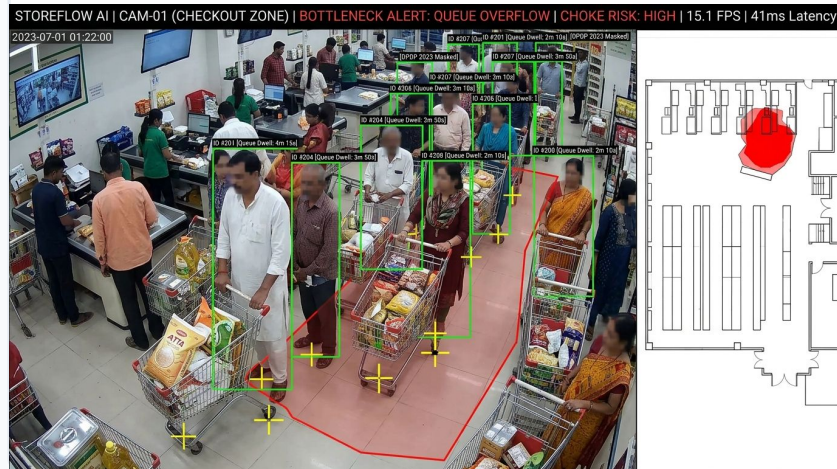
STAGE 03 // MARKOV CHAIN DEAD-ZONE DISCOVERY

Transition Matrix Modeling: Computes Markov probability paths to derive Spatial Opportunity Scores.

Blind Spot Detection: Pinpoints specific aisles with **< 8% discovery rate**.

Actionable Output: Directly highlights flow drop-off points & merchandising dead-zones.

2D CAD FLOOR PLAN // MICRO-CELL DENSITY & CHOKE POINT



STORE FRICTION

78/100

- Flow: Optimal
- Aisle 10: Dwell
- Exit: Bottleneck

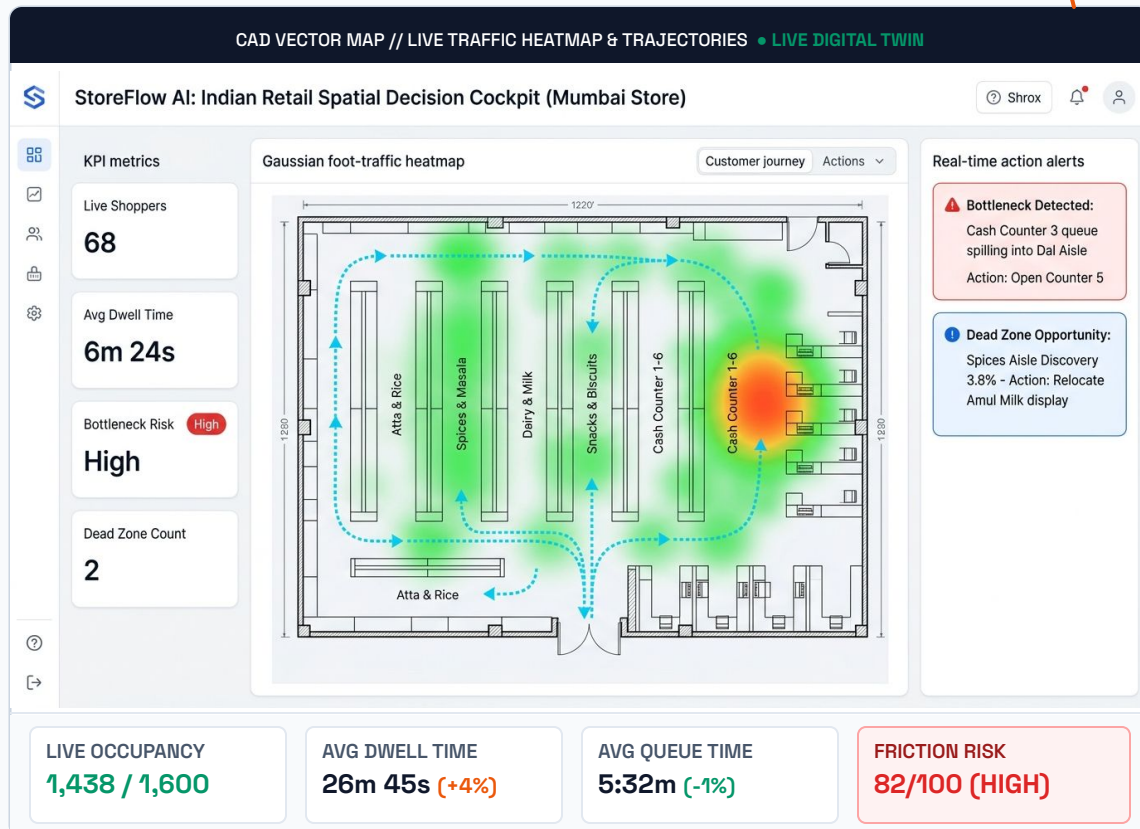
Red choke hotspot flags checkout queue spillover in < 35s!

MST orientation isolates queue accumulation from produce browsing.

The Executive Cockpit: From Raw Feeds to Actionable Store Maps

✓ StoreFlow AI

Instant Action!



01 | MOBILE DISPATCH

30-Sec Mobile Dispatch

WhatsApp/Telegram bottleneck alerts: 'Cash Counter 3 spilling into Dal Aisle. Open Counter 5.'

02 | CAD VECTORIZATION

Automated Floor Vectorization

Ingests CAD DXF, SVG, or PDFs to build 2D interactive maps with semantic zoning (Dairy, Cash Counters).

03 | TRAFFIC ANALYTICS

Gaussian Heatmaps & Traces

Emerald green browsing glows vs red choke points overlaid with cyan-blue customer trajectory lines.

04 | EXPERIMENTATION ENGINE

Physical A/B Verification

Generates automated Before-vs-After verification reports proving ROI on merchandising changes.



How We Propose & Deploy: The 4-Step Zero-Disruption Supermarket Onboarding Plan



StoreFlow AI

StoreFlow AI is designed for commercial feasibility, transitioning from a paper blueprint to live operational intelligence in under 48 hours with zero downtime.

🕒 STEP 1 • 30 MINS

Ingest Blueprint & Mobile Snap-to-CAD Parsing

Users can upload a CAD/PDF or simply snap a smartphone photo of the fire-exit map. The system performs automated polygon vectorization and semantic department tagging, saving approximately \$3,000 in surveying fees.

🕒 STEP 2 • 15 MINS

Auto-Discover CCTV & Register Camera Frustums

Multicast ONVIF WS-Discovery finds 12–16 cameras in under 45 seconds. A 4-landmark alignment calculates the homography matrix, mapping 3D camera frustums with zero ladders or ceiling wiring required.

🕒 STEP 3 • 10 MINS

Plug-In Edge Appliance to PoE Switch

A single Ethernet cable connects the 12W appliance (costing just ₹100/mo in power) to the CCTV switch. It uses a passive video tap with zero inbound firewall ports, emitting only 3.25 KB/s outbound telemetry.

🕒 STEP 4 • 48 HOURS

48-Hour Baseline Learning & Manager Go-Live

The system conducts unsupervised velocity baselines and POS receipt revenue fusion, activating live mobile WhatsApp alerts within two days.

✓ ZERO DOWNTIME GUARANTEE

Non-Invasive Network Deployment: Operates via passive video tap with zero inbound firewall ports, no ceiling rewiring, and minimal bandwidth consumption (3.25 KB/s).

→ *Live in under 48 hours with zero store disruption!*



Enterprise Viability: >97% Cost Reduction & DPDP 2023 Compliance



StoreFlow AI

HYBRID B2B SAAS MODEL (THIN-MARGIN RETAIL)

Hardware CAPEX: ₹15k-₹35k (\$180-\$420) one-time

Software SaaS: ₹4,999/mo (\$60/mo) • **Payback < 3 Weeks**

COST & TCO COMPARISON (6 DIMENSIONS)

Dimension	Legacy Cloud	StoreFlow AI
Camera CAPEX	\$9,600 (New IP)	\$0 (Existing)
Edge Compute	\$6,500 (Servers)	\$130 (Micro-Node)
Power / Store	750W Continuous	12W (98% Drop)
Bandwidth	32 Mbps Stream	3.25 KB/s (14k x)
Privacy Ops	High Risk / Audit	Zero-PII Native
Year 1 TCO	\$24,000 - \$35,000	\$601 - \$969 >97% CUT

➤ *TCO drops from \$35k to under \$1k per store while turning existing CCTV into real-time retail intelligence!*



STATUTORY PRIVACY COMPLIANCE (ZERO-PII)

Fully Aligned with India DPDP Act 2023 & EU GDPR Article 9

1. ZERO VIDEO SAVED TO DISK

Volatile RAM-only processing pipeline. Video frames are immediately processed in-memory and overwritten in real time. Zero local or cloud video storage.

2. ZERO FACIAL BIOMETRICS EXTRACTED

No facial recognition or identity vectorization. Spatial telemetry tracks anonymized bounding boxes and body pose keypoints without biological identity data.

3. EPHEMERAL SESSION TOKENS (30-MIN TTL)

Temporary operational identifiers automatically expire after 30 minutes. Prevents cross-day or multi-store tracking of shoppers.

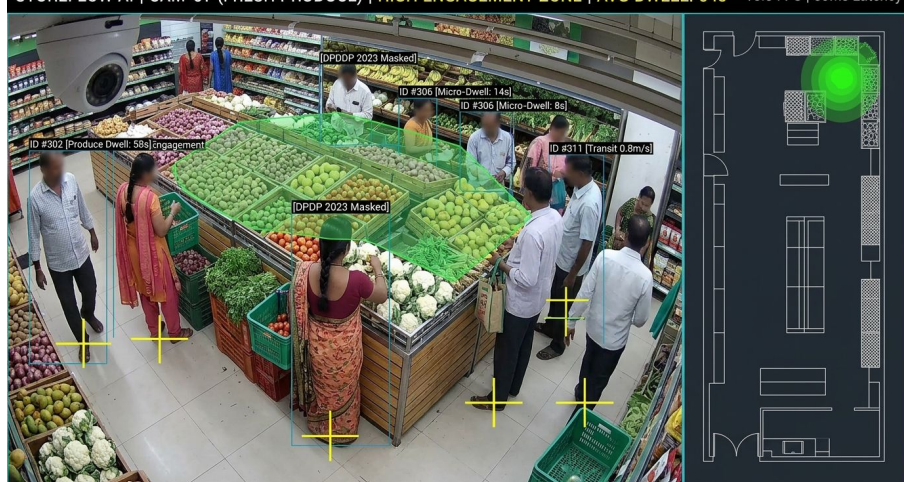
4. AUTOMATED EMPLOYEE FILTERING

Uniform HSV chromatic clustering automatically decouples staff movements from customer dwell metrics, ensuring 100% pure shopper analytics.

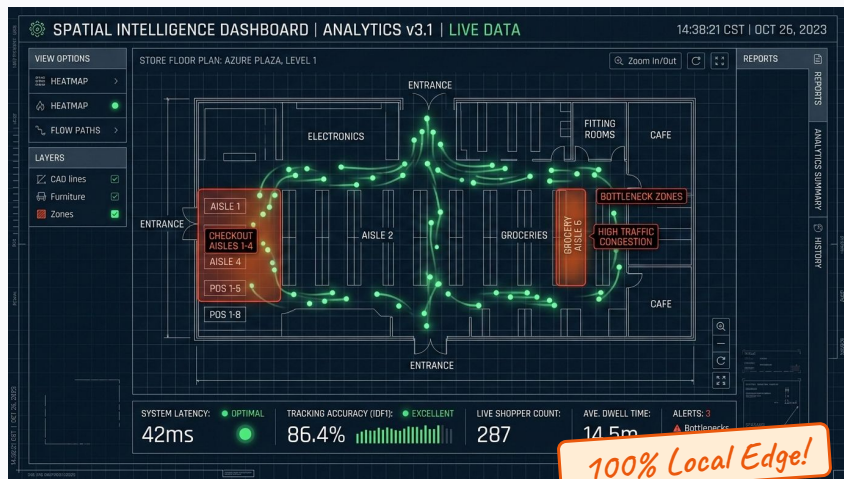
STOREFLOW AI | CAM-02 (ENTRANCE & PROMO ISLAND) | EPR YIELD: 32.4% | 15.2 FPS | 40ms Latency



STOREFLOW AI | CAM-07 (FRESH PRODUCE) | HIGH ENGAGEMENT ZONE | AVG DWELL: 54s 15.0 FPS | 39ms Latency



Target Engineering Benchmarks, Live WebGL PoC & Commercial Rollout



TRIPLE-PROOF LIVE PITCH DEMONSTRATION

- **Local Edge PoC:** 100% self-contained execution with zero internet required.
- **Hardware Performance:** 42ms on Jetson Orin Nano / 78ms on Intel N100 (8 streams).
- **Active Telemetry:** Split-screen PoC with interactive 35s bottleneck lead time alerts.

TARGET ENGINEERING BENCHMARKS

86.4%

IDF1 TRACKING (MOT20)

$\pm 0.11m$

SPATIAL ACCURACY (RMSE)

42ms

EDGE PIPELINE LATENCY

93.2%

DWELL TIME ACCURACY

90-DAY COMMERCIAL DEPLOYMENT ROADMAP

PHASE 1

Days 1–30: Plug-and-Play Pilot

Initiate edge box plug-and-play pilot in 3 test store locations.

PHASE 2

Days 31–60: POS Fusion & Auto-Dispatch

Integrate checkout POS fusion and automated staff dispatch.

PHASE 3

Days 61–90: Enterprise Chain Rollout

Scale to enterprise chain rollout targeting **+3–7% top-line sales lift** and an **18% queue wait reduction**.